Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Data Analysis using SQL (Business Transactions)

Analysis Queries

1. Total Revenue Generated by Male vs Female Customers

```
SELECT Gender,
SUM(Purchase_Amount_USD) AS Revenue
FROM customer_shopping_behavior_analysis.customer_shopping_behavior
GROUP BY Gender;
```

gender	revenue
Male	157890
Female	75191

Meaning

- SELECT Gender → shows male and female

- SUM(Purchase_Amount_USD) → calculates total revenue
- GROUP BY Gender → separates revenue by gender

2. Customers Who Used Discount and Spent Above Average

```
SELECT Customer_ID,
Purchase_Amount_USD
FROM customer_shopping_behavior_analysis.customer_shopping_behavior
WHERE Discount_Applied = 'Yes'
AND Purchase_Amount_USD >= (
  SELECT AVG(Purchase_Amount_USD)
  FROM customer_shopping_behavior_analysis.customer_shopping_behavior
```

customer_id	purchase_amount_usd
2	64
3	73
4	90
7	85
9	97
12	68

Meaning

- WHERE Discount_Applied = 'Yes' → customers who used discounts
- AVG() → calculates average purchase amount
- >= → finds customers spending above average

3. Top 5 Products with Highest Average Review Rating

```
SELECT Item_Purchased,
AVG(Review_Rating) AS Average_Product_Rating
FROM customer_shopping_behavior_analysis.customer_shopping_behavior
GROUP BY Item_Purchased
ORDER BY AVG(Review_Rating) DESC
LIMIT 5;
```

	item_purchased	average product rating
▶	Gloves	3.860357142857143
	Sandals	3.8440624999999997
	Boots	3.818402777777778
	Hat	3.8009740259740252
	Skirt	3.7851265822784 8

Meaning

- AVG(Review_Rating) → average rating of products
- ORDER BY DESC → highest ratings first
- LIMIT 5 → top 5 products only

4. Compare Average Purchase Amount Between Standard and Express Shipping

SELECT Shipping_Type,
ROUND(AVG(Purchase_Amount_USD), 2) AS Avg_Purchase
FROM customer_shopping_behavior_analysis.customer_shopping_behavior
WHERE Shipping_Type IN ('Standard', 'Express')
GROUP BY Shipping_Type;

	shipping_type	round(avg(purchase_amount_usd),2)
▶	Express	60.48
	Standard	58.46

Meaning

- IN() → filters Standard and Express shipping
- AVG() → average purchase amount
- ROUND(...,2) → shows 2 decimal values

5. Compare Subscribers vs Non-Subscribers Spending

SELECT Subscription_Status,
COUNT(Customer_ID) AS Total_Customers,
ROUND(AVG(Purchase_Amount_USD), 2) AS Avg_Spend,
ROUND(SUM(Purchase_Amount_USD), 2) AS Total_Revenue
FROM customer_shopping_behavior_analysis.customer_shopping_behavior

GROUP BY Subscription_Status
ORDER BY Total_Revenue DESC;

	Subscription_Status	total_customers	avg_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

Meaning

- COUNT() → total customers
- AVG() → average spending
- SUM() → total revenue
- GROUP BY Subscription_Status → subscriber vs non-subscriber

6. Top 5 Products with Highest Discount Purchase Percentage

SELECT Item_Purchased,ROUND((SUM(CASE WHEN Discount_Applied = 'Yes' THEN 1 ELSE 0 END) * 100.0) / COUNT(*), 2) AS Discount_PercentageFROM customer_shopping_behavior_analysis.customer_shopping_behavior

GROUP BY Item_Purchased

ORDER BY Discount_Percentage DESC

LIMIT 5;

	Item_Purchased	Discount_Percentage
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Meaning

- CASE WHEN → checks discount applied
- SUM() → counts discount purchases
- COUNT(*) → total purchases
- Formula calculates percentage

7. Segment Customers into New, Returning, and Loyal

```
SELECT
CASE
   WHEN Previous_Purchases = 1 THEN 'New Customer'
   WHEN Previous_Purchases BETWEEN 2 AND 10 THEN 'Returning Customer'
   ELSE 'Loyal Customer'
END AS Customer_Segment,
COUNT(*) AS Total_Customers
FROM customer_shopping_behavior_analysis.customer_shopping_behavior
GROUP BY Customer_Segment;
```

Customer_Segment	Total_Customers
Loyal Customer	3116
Returning Customer	[illegible]
New Customer	83

Meaning

- CASE → creates customer categories
- BETWEEN → checks range
- COUNT(*) → counts customers in each segment

8. Top 3 Most Purchased Products Within Each Category

```
SELECT Category,
Item_Purchased,
COUNT(*) AS Total_Purchases
FROM customer_shopping_behavior_analysis.customer_shopping_behavior
GROUP BY Category, Item_Purchased
ORDER BY Category, Total_Purchases DESC
LIMIT 3;
```

Category	Item_Purchased	Total_Purchases
Accessories	Jewelry	171
Accessories	Sunglasses	161
Accessories	Belt	161

Meaning

- COUNT(*) → total purchases
- GROUP BY → category-wise product grouping
- ORDER BY DESC → highest purchased first

9. Are Repeat Buyers More Likely to Subscribe?

```
SELECT Subscription_Status,
COUNT(Customer_ID) AS Repeat_Buyers
FROM customer_shopping_behavior_analysis.customer_shopping_behavior
WHERE Previous_Purchases > 5
GROUP BY Subscription_Status;
```

subscription_status	repeat_buyers
Yes	958
No	2518

Meaning

- WHERE Previous_Purchases > 5 → repeat buyers
- COUNT() → total repeat buyers
- GROUP BY Subscription_Status → subscriber comparison

10. Revenue Contribution of Each Age Group

```
SELECT Age,
SUM(Purchase_Amount_USD) AS Total_Revenue
FROM customer_shopping_behavior_analysis.customer_shopping_behavior
GROUP BY Age
ORDER BY Total_Revenue DESC;
```

age	total_revenue
49	5552
69	5484
25	5372
41	5282
54	5282
57	5200

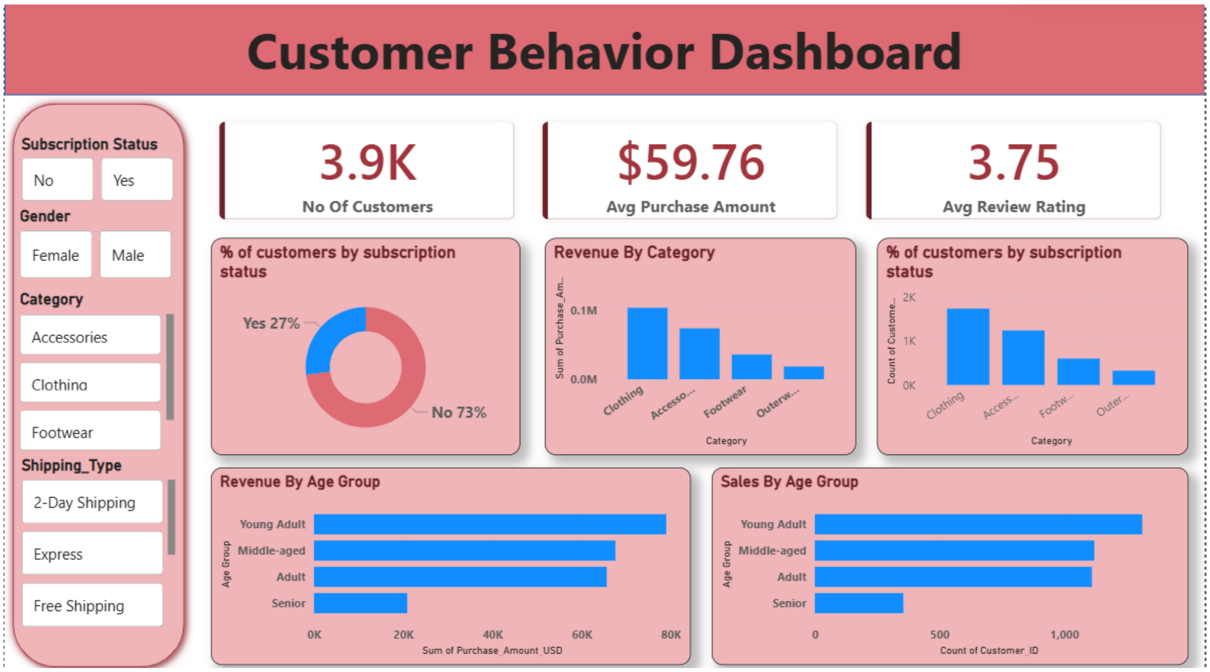
Result 10

Meaning

- SUM() → total revenue
- GROUP BY Age → revenue by age
- ORDER BY DESC → highest revenue first

4. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



5. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.